CHAPTER 11

Orthogonality and distances

S48, S51, S54, S56, S60

Before you begin · Scalar product and triangles

Without a calculator, in six minutes. Answer briefly. Use Pythagoras, angles and coordinates.
Then choose the targeted review indicated below the questions.

1. Calculate $(1;2)\cdot(3;-1)$.
2. Which scalar product characterises two orthogonal vectors?
3. State Pythagoras.
4. Which unit is used for an area?
5. In an orthonormal coordinate system, calculate the norm of $(3;4)$.
6. Give the cosine of a right angle.

Answers

1 ; 0 ; $c^2 = a^2 + b^2$ in a right-angled triangle; one unit squared; 5; 0.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Multiply the corresponding coordinates, then add; R60. **2.** Go back to the zero scalar product criterion in the plane. **3.** First identify the hypotenuse in a right triangle. **4.** Count the unit squares in a rectangle. **5.** Apply Pythagoras to both coordinates. **6.** Read the right-angle point on the circle; R85. The R questions serve as review prompts with the teacher, even if their collective routine is planned for later.

Objectives

Calculate a scalar product; justify orthogonality; find a projection and a distance; calculate a volume.

Course notes

The coordinate calculations in this chapter are done in an orthonormal coordinate system of space.

Scalar product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u}\cdot\vec{v} = \|\vec{u}\|\|\vec{v}\|\cos\theta$ with $\theta \in [0,\pi]$ is their angle. If one is zero, the scalar product is 0.

$$\vec{u}\cdot\vec{v} = \tfrac{1}{2}(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal coordinate system, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The scalar product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their scalar product is zero.

Example. $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of this plane. A vector normal to a plane is a non-zero vector orthogonal to all its directions.

Why two directions are enough. If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

Vector condition for membership

If A lies in the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal projection H of M onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane ; if M lies in the plane, $H = M$. The distance from M to the plane is MH. On a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Proof of the minimum. For every point P of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, therefore $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the point of the plane closest to it.

Method on a line. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Thus $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}}\, h}{3}$, & h is an orthogonal distance. Units are squared and cubed respectively.

Expansions and loci

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

We can therefore also calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two distinct points A, B w/ midpoint I, the locus of points M such that $MA = MB$ is the plane passing through I w/ normal $\overrightarrow{AB}$: the *plan* bisector . Indeed, writing the vectors from I, expanding gives $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$. **Extension, with no dedicated assessment in 2026-27.** Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallel planes, which correspond to collinear normals.

See a projection, then calculate it

11.1 A scalar product may be negative

First use: S48.

Definition · Scalar product

For two non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$ with $\theta \in [0;\pi]$ is their angle. In a orthonormal coordinate system, this product is $xx' + yy' + zz'$. A zero value characterises the orthogonality of directions.

Worked example · Recognising an obtuse angle

For $\vec{u} = (1;0;0)$ and $\vec{v} = (-1;1;0)$, the product is -1, the norms are 1 and $\sqrt{2}$, therefore $\cos\theta = -1/\sqrt{2}$ and $\theta = 3\pi/4$. A negative product indicates an obtuse or straight angle, not a negative length.

A1 · Immediate application Calculate the angle between $(1;1;0)$ and $(1;0;1)$ in an orthonormal coordinate system. Could the same calculation be used in any basis?

11.2 Two directions are needed

First use: S51.

Property · Criterion for orthogonality to a plane

A non-zero vector is normal to a plane if it is orthogonal to two non-collinear direction vectors of this plane. By bilinearity, it is then orthogonal to any combination of these two vectors, hence to every direction in the plane.

Worked example · Testing the three conditions

For $\vec{u} = (1;1;0)$, $\vec{v} = (0;1;1)$ and $\vec{n} = (1;-1;1)$, $\vec{n} \cdot \vec{u} = 0$ and $\vec{n} \cdot \vec{v} = 0$. The two directions are not collinear and $\vec{n} \neq 0$, therefore $\vec{n}$ is normal to the plane they span.

A2 · Counterexample "Being orthogonal to one line in the plane is enough to be normal to the plane." Disprove using two perpendicular directions in the horizontal plane. Also explain the difference between orthogonal lines and perpendicular lines.

Worked example · A locus of equidistant points

Let A=(-1;0;0) and B=(1;0;0). Their midpoint is O. For M=(x;y;z), $MA^2 - MB^2 = (x+1)^2 - (x-1)^2 = 4x$. Thus $MA = MB$ is equivalent to x=0: the locus is the plane through O, normal to $\overrightarrow{AB}$. It is not just the perpendicular bisector drawn on a single face.

A3 · Application and justification For A=(0;0;0), B=(2;2;0), describe geometrically the locus of the points equidistant from A and B. Use the squares of the distances to give a relation between the coordinates; find two points of this locus.

11.3 The foot is the nearest point

First use: S54 and S55.

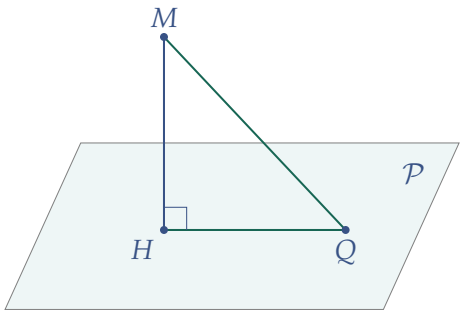


Figure 11.1. The marking shows $MH \perp \mathcal{P}$. For each Q of the plane, $MQ^2 = MH^2 + HQ^2$.

Proof. If H is the orthogonal projection of M onto a plane $\mathcal{P}$, for every $Q \in \mathcal{P}$, the triangle MHQ is right-angled at H. Thus $MQ^2 = MH^2 + HQ^2 \geq MH^2$. Equality requires $HQ = 0$, therefore $Q = H$. The minimum is reached at a single point. The reasoning is the same for a line. □

Worked example · Projecting onto a line without a plane equation

Let $A = (1;0;0)$, $M = (0;2;3)$ and a line through point A with direction $\vec{u} = (1;1;0)$. Write $\overrightarrow{AH} = t\vec{u}$ gives $H = (1+t;t;0)$. The condition $\overrightarrow{MH} \cdot \vec{u} = 0$ becomes $(1+t) + (t-2) = 0$, therefore $t = 1/2$. Thus $H = (3/2;1/2;0)$ and $MH^2 = 27/2$. We check both membership of the line and orthogonality.

A4 · Partial guidance Project $M = (2;3;1)$ onto the line through O with direction $(1;1;0)$. Determine the projection and the distance.

A5 · Choice of strategy Project $M = (1;2;3)$ onto the plane spanned by $\vec{i}, \vec{j}$. Justify geometrically the coordinates of the foot, then prove that every other point of the plane is further away.

11.4 One quantity by two calculations

First use: S56.

Worked example · Recovering an altitude

In an orthonormal coordinate system, $O = (0;0;0)$, $A = (6;0;0)$, $B = (0;4;0)$, $C = (0;0;3)$. The triangle OAB is right-angled with area 12 and OC is perpendicular to its plane; the volume is 12 $\times 3/3 = 12$. If the area of ABC is 3 $\overline{29}$, its altitude from O is 3 $V/\mathcal{A}(ABC) = 12/\overline{29}$.

A6 · Synthesis In this tetrahedron, calculate AB, AC and $\overrightarrow{AB} \cdot \overrightarrow{AC}$. Deduce the cosine and then the sine of the angle $\widehat{BAC}$, check the area of ABC and recover the altitude from O.

A first step: Use $\mathcal{A}(ABC) = \frac{1}{2}$AB AC $\sin \widehat{BAC}$. An angle of a triangle has a positive sine.

A7 · Fix a check A candidate gives a point H of the plane and states "it is the projection since H lies in the plane". What second check is missing? Give two points of the plane $z = 0$ to show that membership alone is not enough.

One minute, spoken aloud. Describe the projection figure, naming the external point, the plane, the foot and the right angle; explain why we do not simply calculate the distance to any point of the plane.

Delayed review · S70. Recalculate the foot with a parametric representation and a Cartesian equation; the two geometric conditions remain the same.

Investigation

The nearest point on a line - 35 min.

In an orthonormal coordinate system of space, consider $A(1;0;0)$, $M(0;2;3)$ and the line d through A, with direction vector $\vec{u} = (1;1;0)$. We look for the point on d closest to M.

1. Express, in terms of a real parameter t, the coordinates of a point H of d.

2. Determine the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the point H thus obtained.

3. Calculate distance MH.

4. Let P be any point of d. Prove that $MP^2 = MH^2 + HP^2$, then deduce that H is the unique point on d closest to M.

Graded exercises

E1 · Direct application. In an orthonormal basis of space, consider $\vec{u} = (2;-1;3)$ and $\vec{v} = (1;4;0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Application and justification. In an orthonormal basis, consider $\vec{u} = (a;2;1)$ and $\vec{v} = (3;-1;4)$with a is real. Determine a so that $\vec{u}$ and $\vec{v}$ are orthogonal.

E3 · Method to organise. In an orthonormal coordinate system, a plane is spanned by $\vec{u} = (1;-1;0)$ and $\vec{v} = (1;0;-1)$. Show that $\vec{n} = (1;1;1)$ is normal to it. We then consider $A(0;0;0)$ and $B(2;2;2)$. Describe geometrically the set of points equidistant from A and B.

E4 · Choice of strategy. In an orthonormal basis, consider $\vec{u} = (1;0;1)$ and $\vec{v} = (0;1;1)$. Determine the measure, between 0 and π, of the angle between these two vectors.

E5 · Synthesis or proof. A tetrahedron's base is a right triangle whose two perpendicular sides measure 3 cm and 4 cm. The height of the tetrahedron relative to this base is 5 cm. Calculate its volume in cm ^{3}.

Work from the textbook or a past paper

Barbazo, p. 362, Exercises 23 and 27. In S48: exercise 23, to calculate a scalar product in coordinates. At home after S54: exercise 27; check orthogonality to the two direction vectors of a plane, without settling for a single zero product.

Homework

DM11 - A face and its altitude - 50 min.

In an orthonormal coordinate system with origin O, consider the points $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Prove that the triangle ABC is equilateral and calculate its area.

2. Show that the vector $\vec{n} = (1;1;1)$ is normal to the plane (ABC).

3. Let H be the orthogonal projection of O on (ABC). Justify that its coordinates are of the form $(t;t;t)$. Determine t using $\overrightarrow{AH}\cdot\vec{n}=0$.

4. Calculate OH, then the volume of the tetrahedron $OABC$ by taking ABC as base. Check this volume by taking OAB as base.

MCQ A · scalar product and orthogonality

Orthonormal coordinate system. After S51; MCQ B after S56.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. $(1;2;-1)\cdot(2;0;3)$ is:

A. 5 B. -1

C. 1 D. -3

2. The norm of $(2;-1;2)$ is:

A. 9 B. 5

C. $\sqrt{5}$ D. 3

3. To prove that a vector is normal to a plane, it is enough to test that it is orthogonal to:

A. two non-collinear directions of the plane B. a direction of the plane

C. two collinear directions of the plane D. a normal vector to the plane

4. Two nonzero vectors with a negative scalar product form an angle that is:

A. acute B. right

C. obtuse D. zero

5. Two orthogonal lines in space:

A. always intersect B. may be non-coplanar

C. are parallel D. coincide

MCQ B · projection, minimum and volume

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The projection of $M(2;3;4)$ onto the plane $z=0$ is:

A. $(0;0;4)$ B. $(2;3;4)$

C. $(0;0;0)$ D. $(2;3;0)$

7. H is the projection of M onto a plane. For every point P of the plane:

A. $MP^2=MH^2+HP^2$ B. $MP=MH+HP$

C. $MH^2=MP^2+HP^2$ D. $MP=HP$

8. A tetrahedron with base area 6 and height 4 has volume:

A. 24 B. 12

C. 8 D. 6

9. For vectors u, v unit vectors such that $u \cdot v = -\frac{1}{2}$, the angle is:

A. $\pi/3$ B. $2\pi/3$

C. $\pi/2$ D. π

10. $A(1;0;0)$, $M(0;2;3)$, and d with direction $(1;1;0)$ through A. The projection of M on d is:

A. $(1;1;0)$ B. $(0;2;0)$

C. $(1;0;0)$ D. $(\frac{3}{2};\frac{1}{2};0)$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every point of a plane is closer to M than its projection.

2. Two vectors with zero scalar product are collinear.

3. Calculating a volume from two different faces must give the same result.

MCQ solutions

1. **B.** $1 \times 2 + 2 \times 0 - 1 \times 3 = -1$. The scalar product is a real number, not a vector.
2. **D.** The sum of the squares is 9; the norm is its square root, that is 3.
3. **A.** Two non-collinear directions generate the direction of the plane. A single test leaves directions not perpendicular to the vector.
4. **C.** The cosine of the angle is negative since the norms are positive. The non-oriented angle is therefore between $\pi/2$ and π.
5. **B.** Orthogonality concerns their directions. Perpendicular lines, moreover, intersect.
6. **D.** The point belongs to the plane and the vector that connects it to M is vertical, so normal to the plane.
7. **A.** The triangle MHP is right-angled at H when $P \neq H$; the identity remains valid for $P = H$. It proves $MP \geq MH$.
8. **C.** The volume is one third of the base area times the height: $6 \times 4/3 = 8$.
9. **B.** The scalar product is the cosine of the angle since the norms are 1; the non-oriented angle is $2\pi/3$.
10. **D.** Write $H = A + t(1;1;0)$. Orthogonality $(M - H) \cdot (1;1;0) = 1 - 2t = 0$ gives $t = 1/2$.

True/false. False: the projection gives the minimum, by Pythagoras. False: $(1;0;0)$ and $(0;1;0)$. True: each height must correspond to the chosen base.

Solutions to the applications and guidance

Application 1. Product 1, norms $\sqrt{2}$ and $\sqrt{2}$, therefore cosine $1/2$, angle $\pi/3$. The coordinate formula requires an orthonormal coordinate system.

Application 2. In the plane $z = 0$, $\vec{\imath} \cdot \vec{\jmath} = 0$, but $\vec{\jmath}$ lies in the plane and is not normal. Lines orthogonal lines have orthogonal directions; they are perpendicular if they also intersect.

Application 4. $H = (t;t;0)$ and $\overrightarrow{MH} \cdot (1;1;0) = 2t - 5 = 0$, therefore $H = (5/2;5/2;0)$. $MH^2 = 1/4 + 1/4 + 1 = 3/2$; distance $\sqrt{6}/2$.

Application 5. $H = (1;2;0)$; $\overrightarrow{HM} = (0;0;3)$ is orthogonal to both directions $\vec{\imath}, \vec{\jmath}$. For $Q = (x;y;0)$, $MQ^2 = (x-1)^2 + (y-2)^2 + 9 \geq 9$, equality only at H.

Application 6. $AB = \sqrt{52}$, $AC = \sqrt{45}$, product 36. The cosine is 36 $/\sqrt{2340}$; the sine is $\overline{2340 - 1296}/\sqrt{2340} = 6\sqrt{29}/\sqrt{2340}$. The area is therefore 3 $\overline{29}$ and the altitude 12/ $\sqrt{29}$.

Application 7. Missing: MH orthogonal to the plane (or $\overrightarrow{MH}$ collinear with a normal). For $M = (1;2;3)$, H=(1;2;0) works but Q=(0;0;0), although in the plane, is not the projection.

Teaching notes

Have students find the altitude in the figure before calculating a volume. For calculating an angle, require checking cos $\theta \in [-1;1]$. Keep an exact expression before approximation.

Application 3. The midpoint is I=(1;1;0) and a normal vector is (2;2;0). The locus is the plane through I with this normal. $MA^2 - MB^2 = 4x + 4y - 8$; hence x+y=2. For example (1;1;0) and (2;0;3) work. They are all points of the plane, with no restriction on z.

Teaching notes and solutions

Guidance for the teacher

Start with the three expressions of the scalar product. Orthogonal lines are not necessarily intersecting: "perpendicular" requires intersection. To prove that a line is orthogonal to a plane, choose two non-collinear directions. The general formula for the point-to-plane distance need not be memorised: prefer the projection. Cartesian equations used in the progression in S54 can be introduced via the vector condition, then formalised in S67.

Solutions to the activity

1. A point H of d is written $H = A + t\vec{u}$, therefore $H(1+t, t, 0)$ for $t \in \mathbb{R}$.

2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The scalar product with $\vec{u} = (1,1,0)$ is $1 + t + t - 2 = 2t - 1$. Set to zero gives $t = \frac{1}{2}$ & $H(\frac{3}{2}, \frac{1}{2}, 0)$.

3. The coordinate system is orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. Hence $MH = \frac{3\sqrt{6}}{2}$.

4. For $P \in d$, $\overrightarrow{HP}$ is collinear with $\vec{u}$ and thus orthogonal to $\overrightarrow{MH}$. Pythagoras' theorem gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The calculated point is indeed the unique point on the line closest to M.

E1. In the orthonormal coordinate system, the scalar product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The scalar product is $3a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to it being zero: $3a + 2 = 0$, therefore $a = -\frac{2}{3}$.

E3. The two direction vectors are non-collinear and both scalar products are zero. The midpoint is $I(1,1,1)$. The required locus is the plane through I and normal to $\overrightarrow{AB} = (2,2,2)$, hence with normal vector $(1,1,1)$.

E4. The two vectors are non-zero, with norms $\sqrt{2}$, and their scalar product is 1. So their angle $\theta \in [0, \pi]$ satisfies $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$, i.e. $\theta = \frac{\pi}{3}$.

E5. The area of the base rectangle is $\frac{3 \times 4}{2} = 6$ cm^2. The altitude perpendicular to this base is 5 cm, so $V = \frac{6 \times 5}{3} = 10$ cm^3.

Solutions to homework DM11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; the three norms equal $2\sqrt{2}$. The triangle is equilateral. Its height is $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.

2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ are not collinear. Their scalar products with $(1,1,1)$ are respectively $-2+2=0$ and $-2+2=0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).

3. The projection of O lies on the line with direction vector $\vec{n}$ through O, therefore $H = (t,t,t)$. Its condition for belonging to the plane is $(t-2) + t + t = 0$, i.e. $t = \frac{2}{3}$. We get $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which satisfies the two conditions for a projection.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking the base ABC, the volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ units of volume. Another base: the triangle OAB is right-angled at O, area $\frac{2 \times 2}{2} = 2$; the line (OC) is perpendicular to the plane (OAB), and $OC = 2$. Thus $V = \frac{2 \times 2}{3} = \frac{4}{3}$.

Solutions to the textbook or past-paper work

Exercise 23. In the given orthonormal coordinate system, the scalar product is the sum of the products of the corresponding coordinates:

$$(-3{,}5)(-4) + 2 \times 5{,}5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two vectors $\vec{u}$ and $\vec{v}$ given span the plane and are not collinear. A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal to the plane.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal to the plane.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single nonzero product is enough to rule out normality.